

Erdős Problem #838

EP-838

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f(n)$ be maximal such that any n points in $\mathbb{R}^2$, with no three on a line, determine at least $f(n)$ different convex subsets. Estimate $f(n)$ - in particular, does there exist a constant c such that $\lim_{n \rightarrow \infty} \frac{\log f(n)}{(\log n)^2} = c$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>